

```
[root@deepu waf]# waf
Waf: Entering directory `./root/Desktop/waf/de'
[1/4] c: main.c -> de/main.c.1.o
[2/4] c: data.c -> de/data.c.2.o
[3/4] c: show.c -> de/show.c.3.o
[4/4] program: de/main.c.1.o de/data.c.2.o de/show.c.3.o -> de/hello
Waf: Leaving directory `./root/Desktop/waf/de'
'build' finished successfully (0.794s)
[root@deepu waf]# waf
Waf: Entering directory `./root/Desktop/waf/de'
Waf: Leaving directory `./root/Desktop/waf/de'
'build' finished successfully (0.011s)
```

Figure 4: Back-to-back execution of `wscript`

start working with Waf. The features of Waf that I have left out are innumerable. For example, Waf can build executables in parallel which, all together, reduce the code generation time. Another stunning (and disturbing) fact regarding Waf is that it offers a Turing complete language as part of its repertoire. In simple terms, a tool can be called 'Turing complete' if and only if it is as powerful as the C programming language—at least theoretically. The command `waf distclean` will delete all the files generated by the command `waf` or `waf build`. This is quite an improvement from the Make utility, with which you need to define the function `clean` independently, to remove the files automatically built by Make.

Finally, to understand the single most important benefit of Waf or any other build automation tool in comparison to manual compilation, execute the command `waf` twice on the terminal. I am sure that the second time, Waf will not do anything at all. All you are going to see is the message shown in Figure 4.

We can conclude that for Waf, 'if there's no modification, then there's no compilation'. In simple terms, Waf will recompile a source file if and only if it is modified. But it is also possible to forcefully recompile all the source files even when no changes are made. This can be achieved by executing the command `waf clean` on the terminal. **END**

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